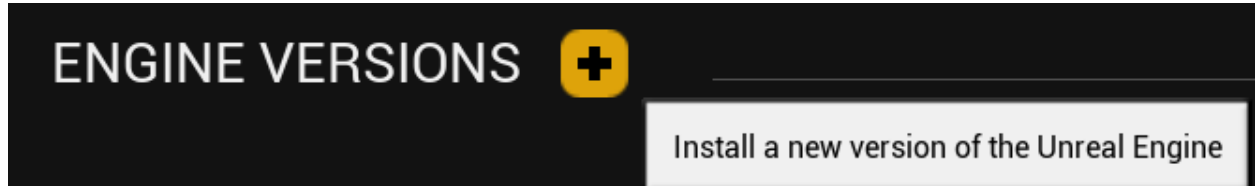


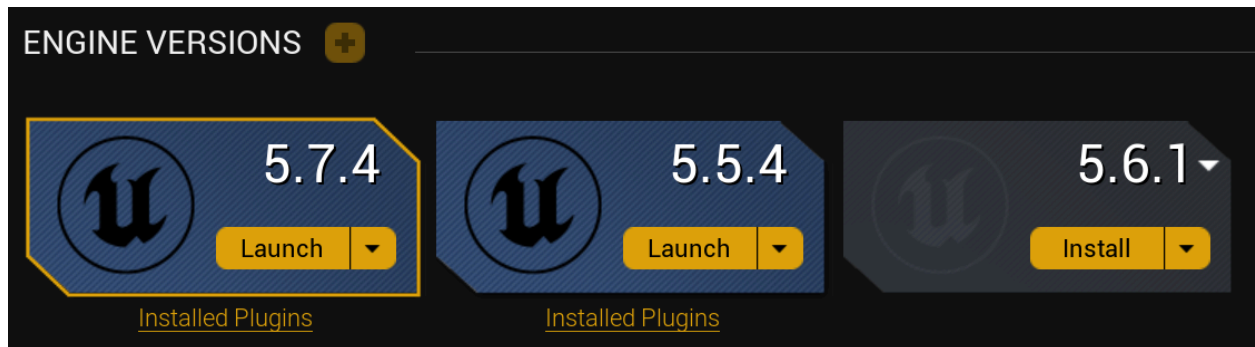
Tutorial: Unreal Engine 5.7.4 Water Material

The first thing we'll need to do to create a realistic water shader is re-import the Starter Content folder. The option to include the Starter Content was removed in UE5.7 and we are going to use a Normal map included inside of it for our water material.

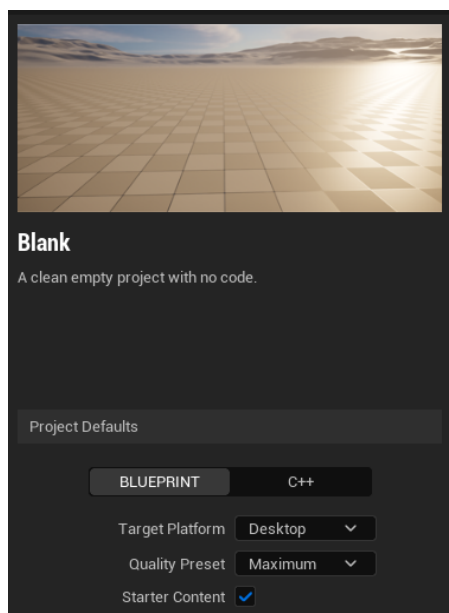
To start, go to your Launcher and install a previous version of Unreal Engine. Versions prior to 5.7 will work, but I did 5.5.4 to be safe. Select the version using the dropdown to the side.



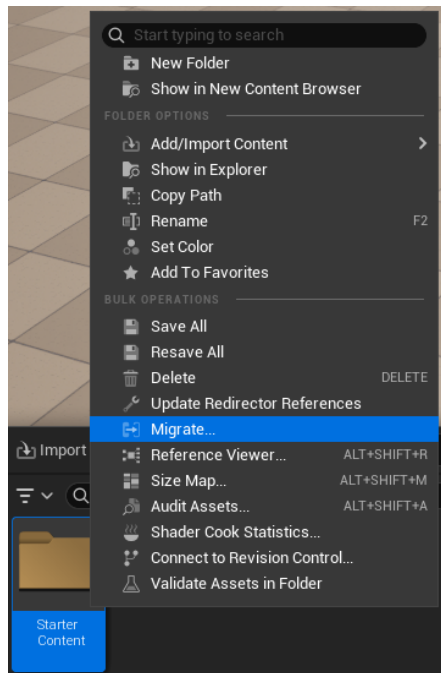
Once the engine version is selected and installed, you can create and launch projects with it.



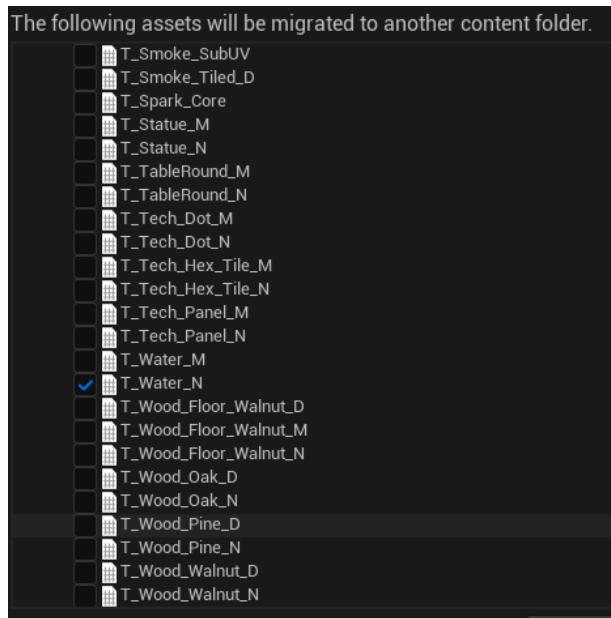
Once you have installed the previous engine version, create a new project and include the Starter Content.



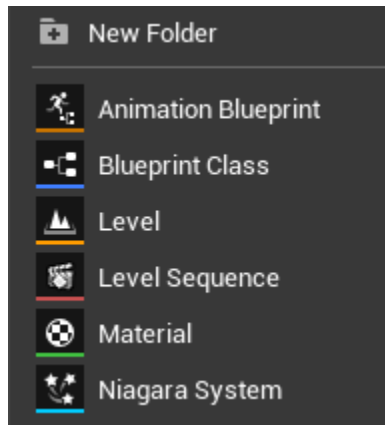
After you load into the project, open the Content Drawer and select the StarterContent folder. Right click it and select Migrate.



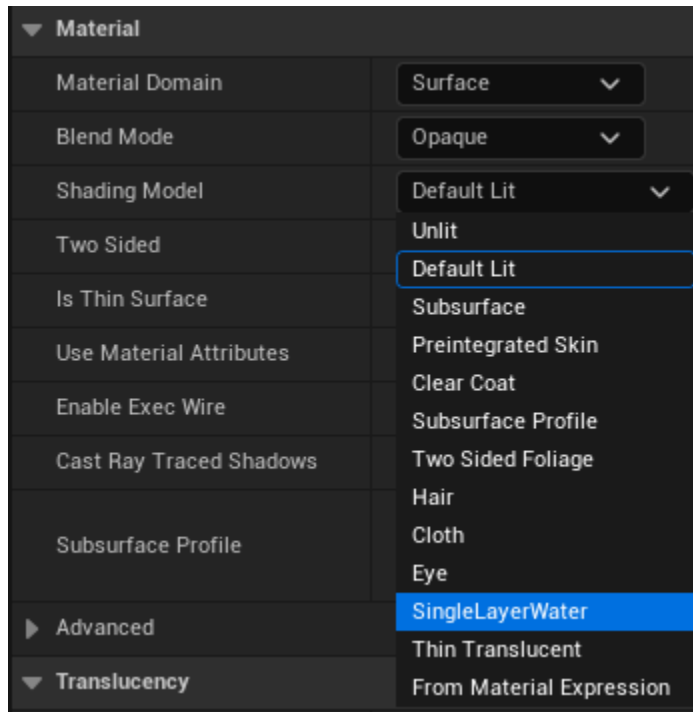
If you want to import the full folder, press OK and select the project you want to import the folder into. However, if you only want to migrate a single asset, you can deselect different fields while importing. Our shader only needs the texture T_Water_N, so you can deselect everything by unchecking StarterContent and only select the texture.



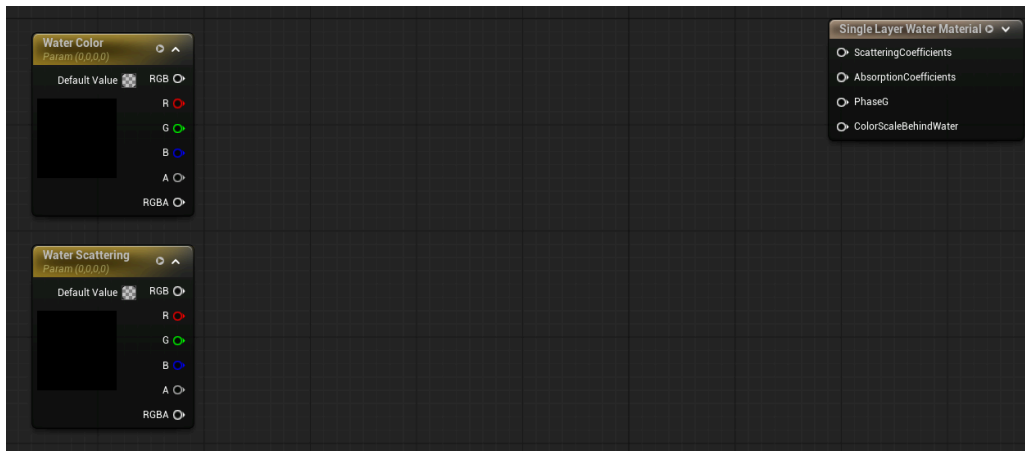
Now we'll start making the water material itself. Reload your UE5.7.4 project and add a Material by right clicking in the Content Drawer and selecting Material.



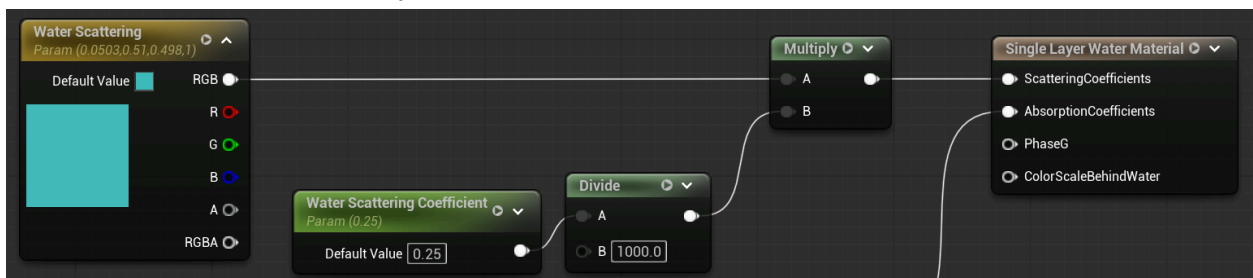
Select the Material and open the Material Editor. Once you have opened it, set the shading model to SingleLayerWater.



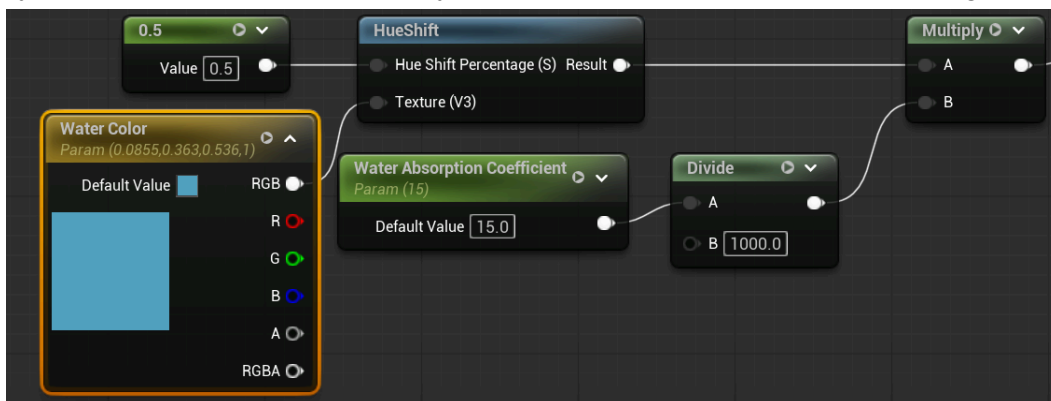
First, we need to set the material's color. Right click and place a SingleLayerWater Material node. We'll use the Scattering and Absorption Coefficients to edit the outputted color. Place down two initial VectorParameter nodes to use for each base color. Name the first node Water Color and the second node Water Scattering.



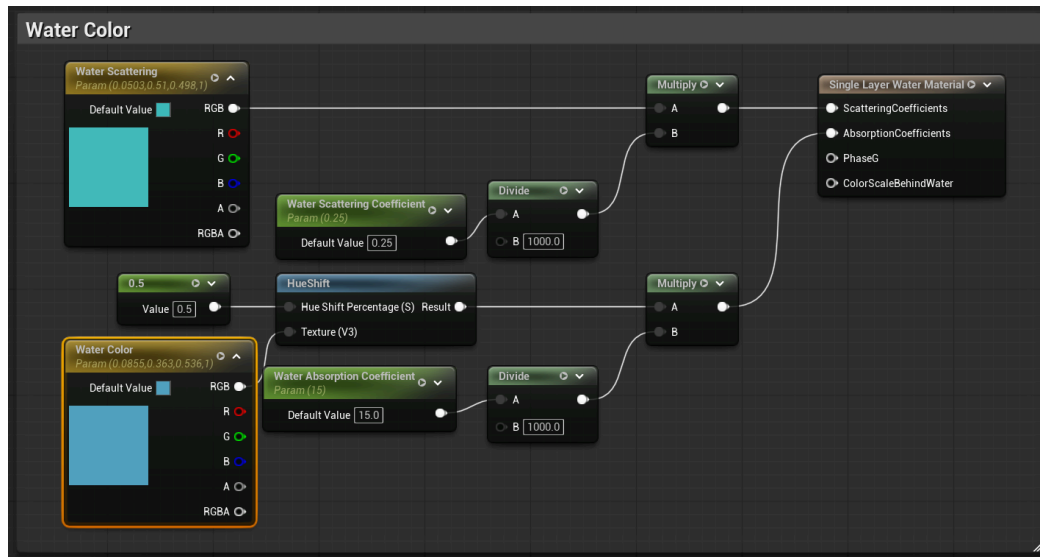
Use a Multiply node to connect the RGB value of your Water Scattering node to ScatteringCoefficients. As the value adjustment is fairly sensitive, we will multiply with a coefficient of 0.25 and divide by 1000.



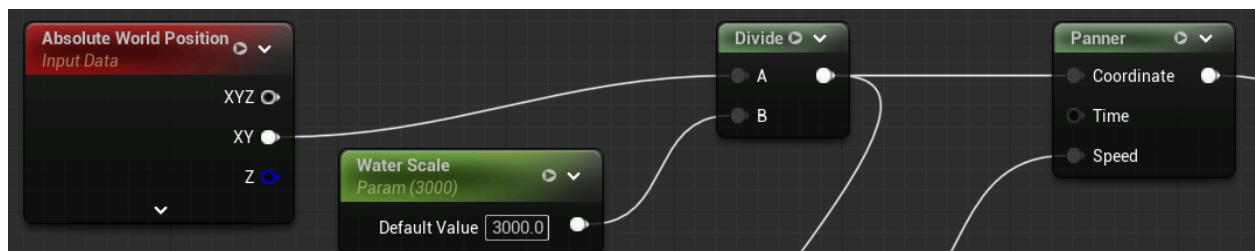
We'll HueShift our Water Color before connecting it to our AbsorptionCoefficients as we want to absorb the opposite hue of the color we specify for our water. Then we multiply the shifted value by a coefficient of 15 and divide by 1000. Coefficients are used so we can group parameters.



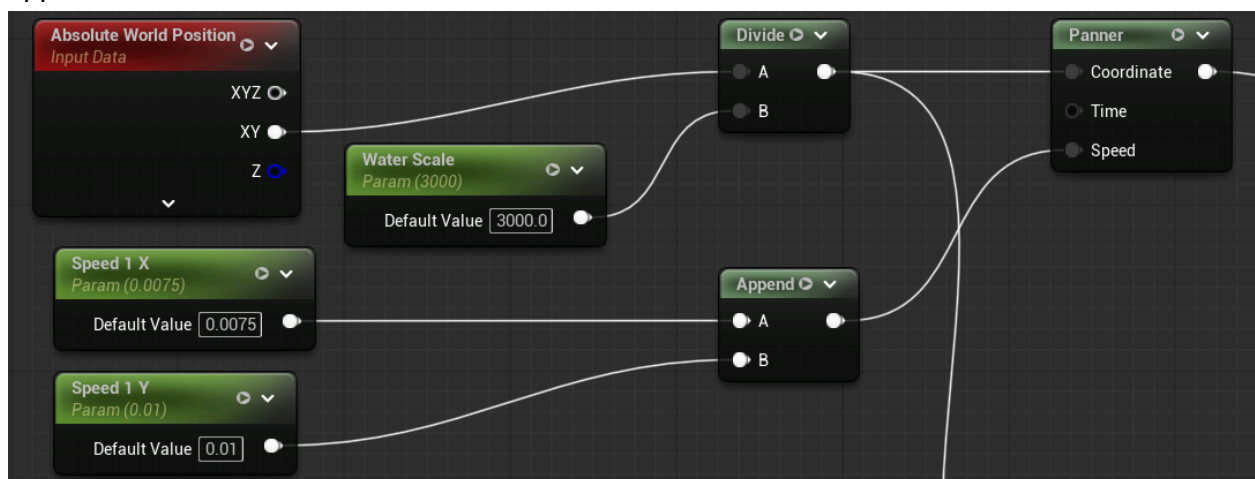
Once the nodes are all connected, select all of them and press C, then label the group Water Color.



Now we'll move on to creating our water's ripple effect. Place down an Absolute World Position node and divide the output by a coefficient of 3000, then place a Panner node and use it as the Coordinate input.



The Speed input on the Panner uses a Vector, so we will need to Append our X and Y panning values together. Create two Parameters labeled Speed 1 X and Y. Combine them using an Append node and connect them to the Panner. Set S1X to 0.0075 and S1Y to 0.01 to start.



The screenshot shows a Pure Data patch for a 3D audio system. The patch is organized into several sections:

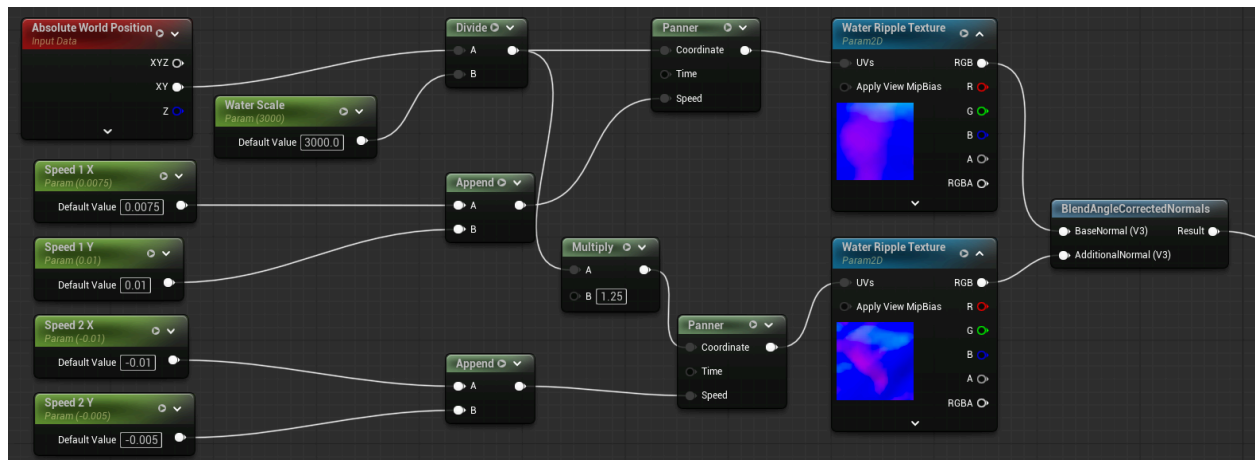
- Input Section:**
 - Absolute World Position:** A module with a dropdown menu and three input fields for XYZ coordinates. The 'Z' input is currently set to 0.
 - Water Scale:** A module with a dropdown menu and a 'Default Value' field set to 3000.0.
 - Speed 1 X:** A module with a dropdown menu and a 'Default Value' field set to 0.0075.
 - Speed 1 Y:** A module with a dropdown menu and a 'Default Value' field set to 0.01.
 - Speed 2 X:** A module with a dropdown menu and a 'Default Value' field set to -0.01.
 - Speed 2 Y:** A module with a dropdown menu and a 'Default Value' field set to -0.005.
- Processing Section:**
 - Divide:** A module that takes the output of the 'Water Scale' module and the 'Speed 1 X' module as inputs.
 - Append:** Two modules that take the outputs of the 'Speed 1 Y', 'Speed 2 X', and 'Speed 2 Y' modules as inputs.
 - Multiply:** A module that takes the output of the first 'Divide' module and a constant value of 1.25 as inputs.
- Output Section:**
 - Panner:** Two modules that take the outputs of the 'Multiply' and 'Append' modules as inputs to control the 3D audio panning.

▼

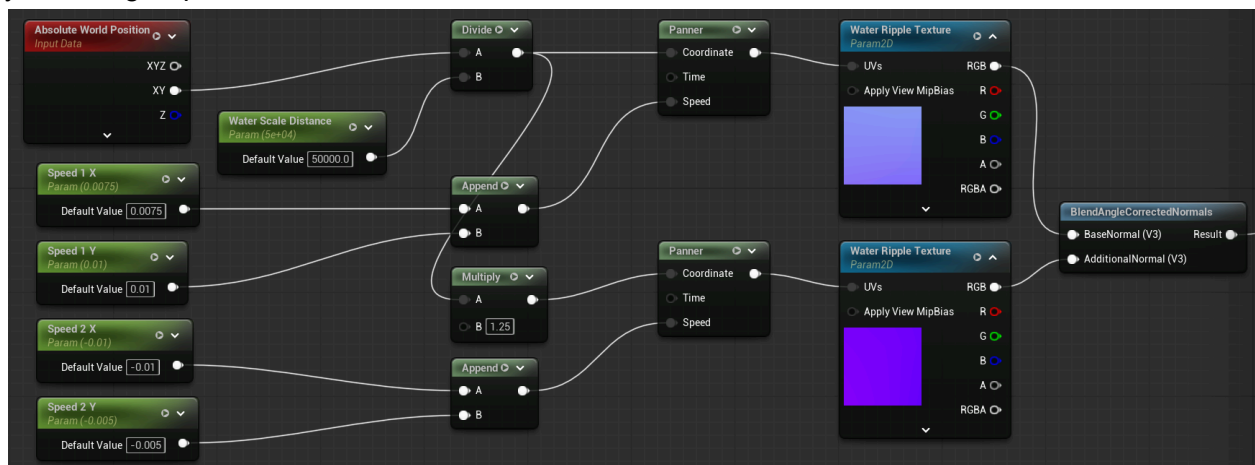
Material Expression Texture Base

Water Ripple Texture		water_n ↶ ↷ 🔍	↶
Sampler Type	Normal		↶
Is Default Meshpaint Texture	☐		

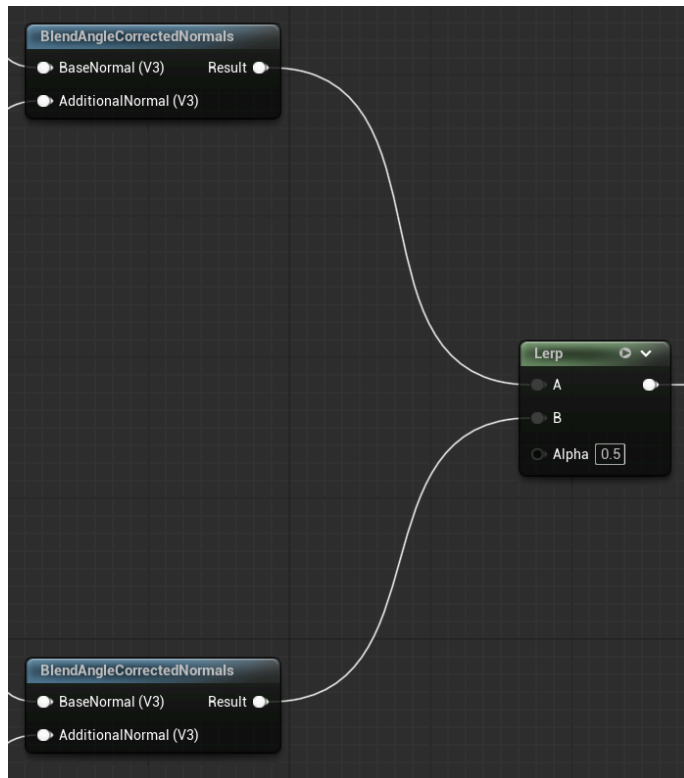
Copy the node and connect one to each of your Panners, then overlap them by using a BlendAngleCorrectedNormals node. This will cause both normal maps to overlay each other and create a composite shimmering effect.



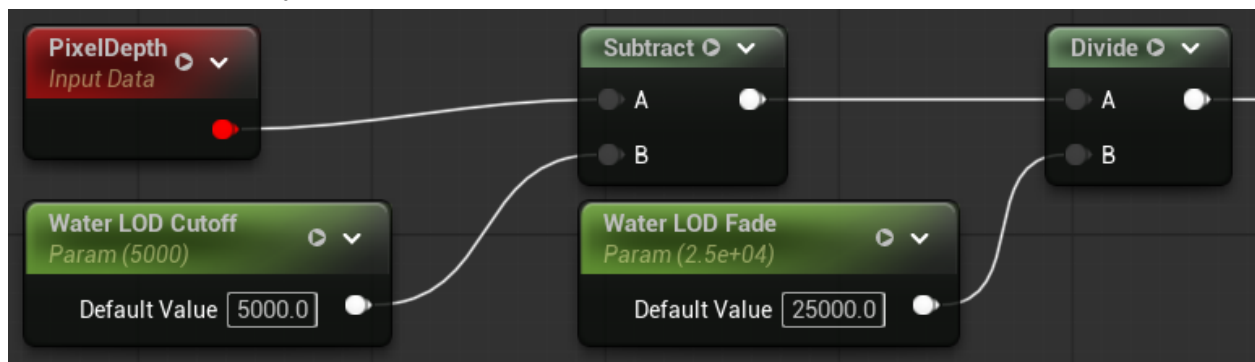
To ensure the material looks good from a wider distance range, we'll blend our initial ripples with a larger set as distance increases. Duplicate the entire WorldPosition group and drag it below your first group, then rename Water Scale to Water Scale Distance and set the value to 50,000.



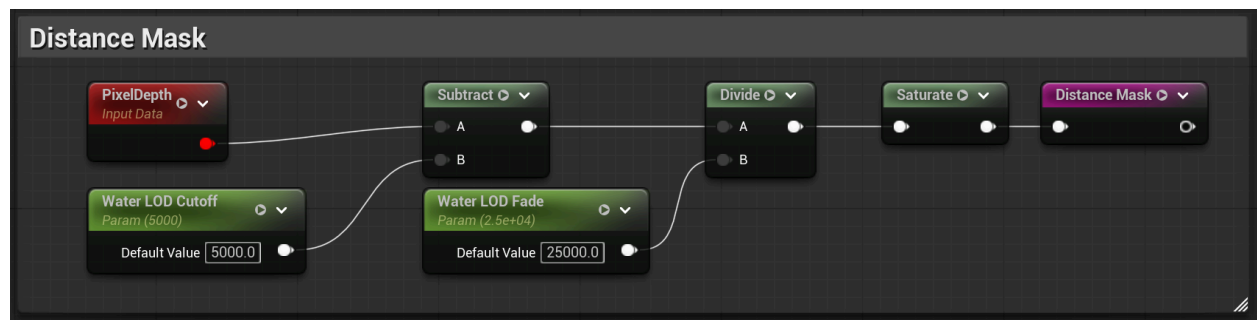
Next, use a Linear Interpolation node and connect both the inputs from your Panners to A and B.



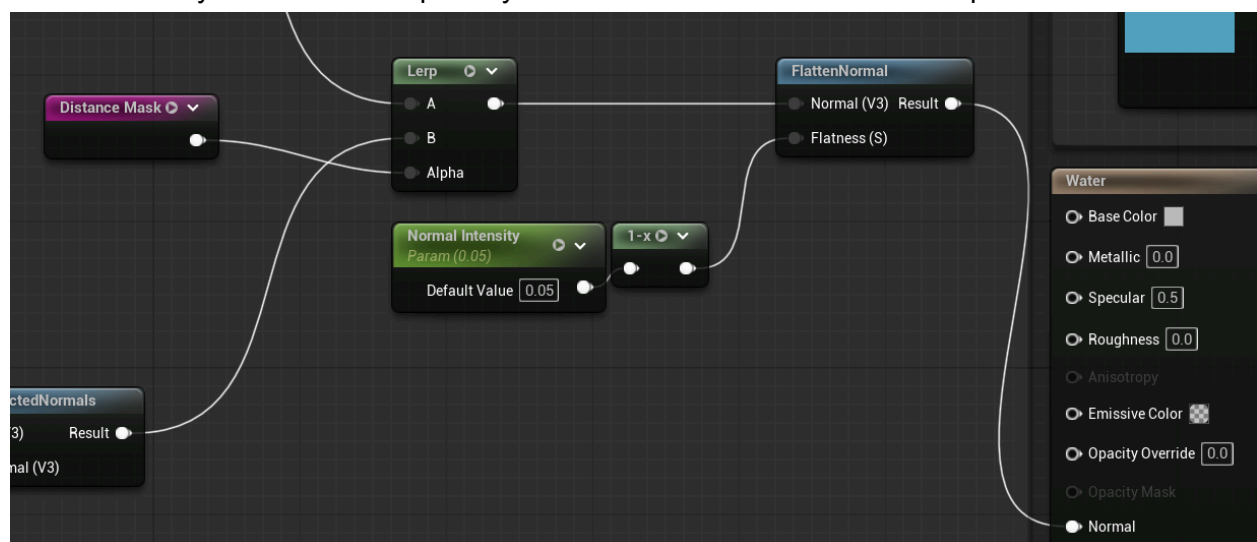
For our Alpha value, we'll create a Distance Mask using PixelDepth to vary our Lerp blending based on distance from the camera. Place down a PixelDepth node, subtract a cutoff of 5000 from it, and divide it by another cutoff of 25000.



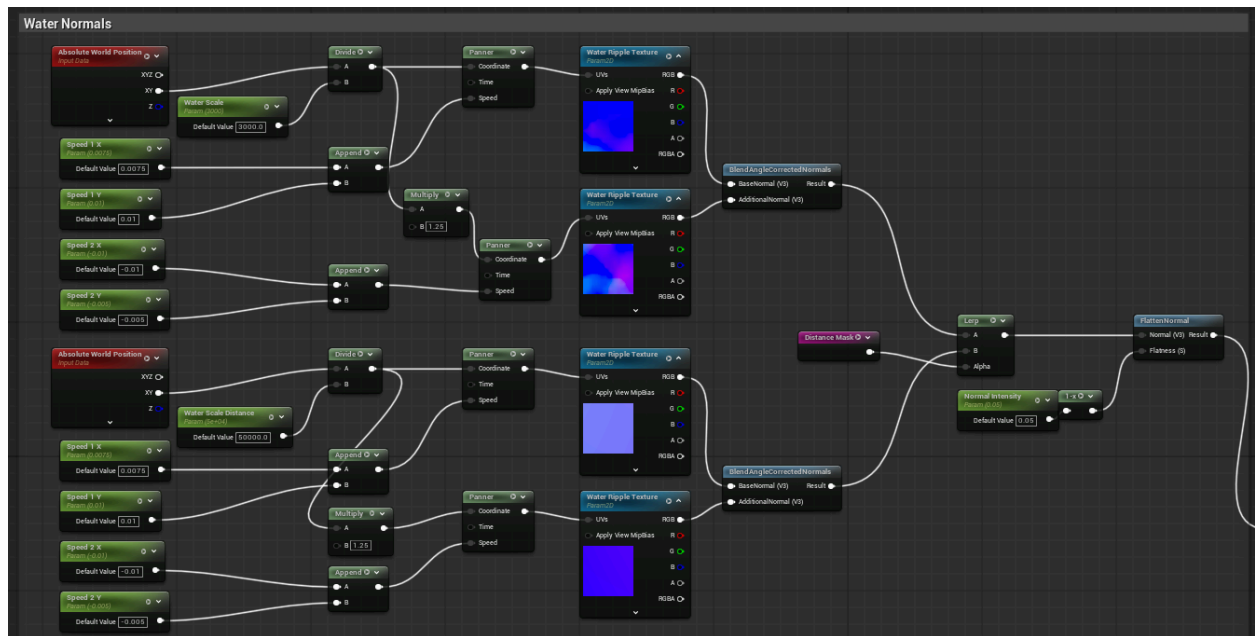
Connect the output from the final Divide to a Saturate node. This clamps it between 0 and 1 and makes it usable as our Alpha value. Then, create a Named Reroute called Distance Mask and connect the output to it. Select these nodes, press C, and label them Distance Mask.



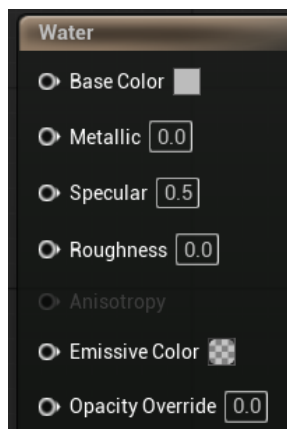
Now that we have a Named Reroute, type Distance and scroll all the way up to select it, then connect it to your Lerp. Connect the Lerp output to a FlattenNormal node and create a Normal Intensity parameter with a value of 0.05. As we want values close to 1, we'll connect this to a 1-X node to get 1-(0.05) or a final value of 0.95 flatness. Connect this to your FlattenNormals node and finally connect the output to your base Material node's Normal input.



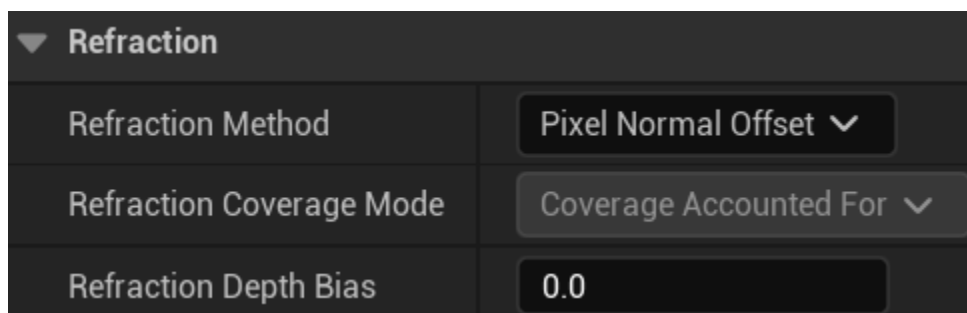
Select the unholy spaghetti monster you just created, press C, and label it Water Normals.



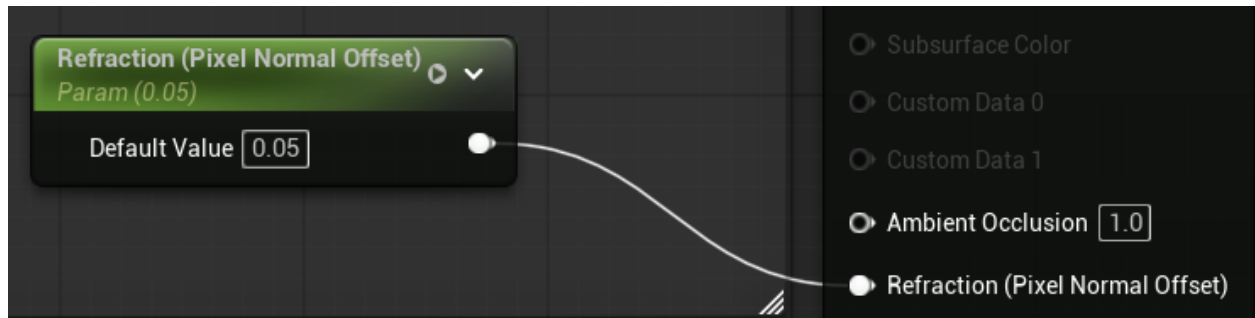
Almost done now! All we need to do is set some base material attributes. On the base Material node, set Metallic to 0, Specular to 0.5, and Roughness to 0. Then, set opacity to 0 as well.



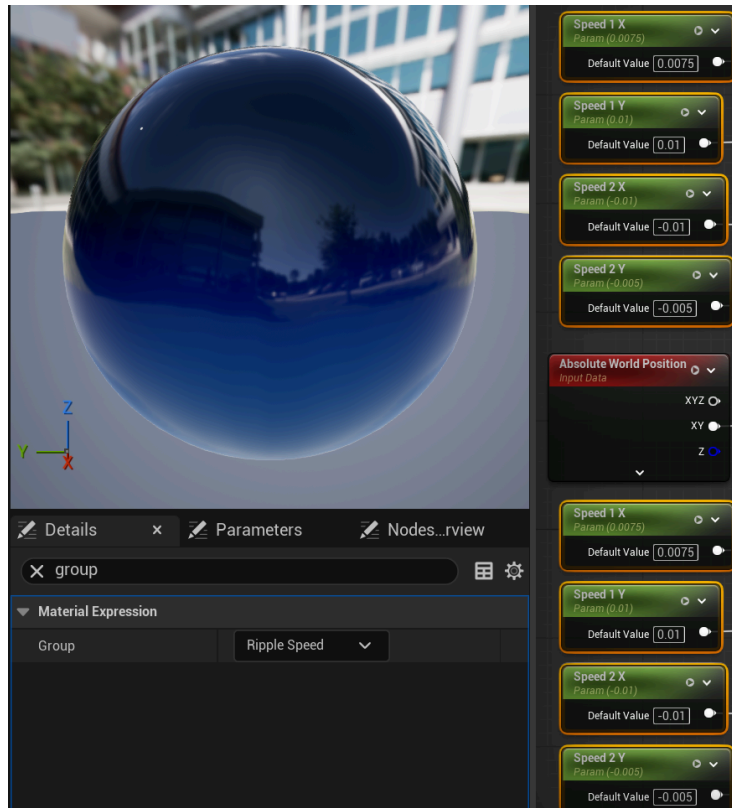
Now we'll enable refraction on our water. Scroll down to Refraction and set the method to Pixel Normal Offset.



Create a new parameter named Refraction and set it to 0.5, connecting it to your Refraction offset. As we want our Refraction value to be grouped, we can't use the in-built offset.



The water material itself is now done! However, I recommend grouping several of your coefficients for easier editing. Start by selecting all your X/Y speed nodes in your Water Normals and adding them to a Ripple Speed group by overwriting “No Group” in the Material Expressions and pressing Enter.



Make 4 more groups for your editable values; first, select your Water Scale coefficients, Normal Intensity, and your Water Ripple Textures and add them to the Ripples group. Next, select your Water LOD Cutoff and Fade values and add them to the Distance group. Select your Water Scattering and Color nodes as well as their coefficients and add them to the Color group, and add your Refraction node to a separate Refraction group.

Parameter Group Data	5 Array elements	
▼ Index [0]	2 members	↶
Group Name	Refraction	
Group Sort Priority	0	↶
▼ Index [1]	2 members	↶
Group Name	Color	
Group Sort Priority	0	↶
▼ Index [2]	2 members	↶
Group Name	Ripples	
Group Sort Priority	0	↶
▼ Index [3]	2 members	↶
Group Name	Ripple Speed	
Group Sort Priority	0	↶
▼ Index [4]	2 members	↶
Group Name	Distance	
Group Sort Priority	0	↶

▼ GLOBAL COLOR	
Water Absor...	15.0
▶ Water Color	
▶ Water Scatte...	
Water Scatte...	0.25
▼ GLOBAL DISTANCE	
Water LOD C...	5000.0
Water LOD F...	25000.0
▼ GLOBAL REFRACTION	
Refraction (Pi...	0.05
▼ GLOBAL RIPPLES	
Speed 1 X	0.0075
Speed 1 Y	0.01
Speed 2 X	-0.01
Speed 2 Y	-0.005
▼ GLOBAL RIPPLES	
Normal Inten...	0.05
Water Ripple...	 water_n ▼
Water Scale	3000.0
Water Scale...	50000.0

That's it! Now that your values are grouped, you can now apply your Material to objects and edit the material's attributes on the fly to tune your water VFX to your liking.